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PROGRAM

Master of Science in Data Science

STUDENT

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COURSE

MSDS 451, Section 001

CLASS

Financial Engineering

TERM

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ASSIGNMENT

Programming Assignment 1

DATE

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Predicting the next-day direction of WEGE3.SA

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1. Problem

The task is to predict the direction of the next session's close-to-close log return. The series is WEGE3.SA. It is not the WTI series used in the course example.

WEG S.A. makes electric motors, generators, transformers, and industrial automation equipment. Its headquarters are in Jaraguá do Sul, in the state of Santa Catarina, Brazil. Its common share is listed on B3. B3 is the stock exchange of Brazil.

On day t the label is 1 when the close-to-close log return is positive, and 0 when the close is flat or down:

$$y_t = \mathbf{1}\{\log(C_t/C_{t-1}) > 0\}.$$

A forecast for day t may use only prices and volume known at the close of day $t-1$. The open, high, low, close, and volume of day t itself are not predictors.

The course example fits an XGBoost classifier and then scores that fit on the same rows used to train it. This report keeps a later block of sessions out of feature selection and out of the hyperparameter search. The finished classifier is scored on that block once.

2. Data and research design

Daily open, high, low, close, and volume come from Yahoo Finance for WEGE3.SA. Yahoo applies the split and dividend adjustment. The request covers 2000-01-01 through 2026-10-11. Yahoo treats the end date as exclusive, so the last session in the file is 2026-10-09. The download has 6,721 sessions.

The program is the numbered scripts in pipeline/. Shared settings are in 01_config.py. The later scripts download prices, build 25 candidate columns, cut the sample, screen the candidates for drift, select columns by an out-of-sample forward search with a noise gate, tune XGBoost, score the hold-out once, and render this report. Python cannot import a module whose name starts with a digit, so 10_run.py loads each file by name.

A row enters the feature table only when all 25 candidates are defined. That table runs from 2007-07-03 through 2026-10-09 and has 4,418 sessions. It is then cut with two settings in 01_config.py.

DEV_END is 2023-12-31. Development is every feature row on or before that date: 3,723 sessions, from 2007-07-03 through 2023-12-28. The next 10 sessions, 2024-01-02 through 2024-01-15, are a gap. They are neither fit nor scored. The remaining 685 sessions, 2024-01-16 through 2026-10-09, are the hold-out.

The drift screen, feature selection, and the hyperparameter search read the development file only. After those choices are fixed, XGBoost is fit on all 3,723 development sessions and applied once to the hold-out. No layer before the hold-out score reads the hold-out file.

3. Features

The target is the direction of the close-to-close log return on WEGE3.SA. On day t,

$$r_t = \log(C_t / C_{t-1}), \qquad y_t = \mathbf{1}\{r_t > 0\}.$$

Each of the 25 candidates is known at the close of day $t-1$. The open, high, low, close, and volume of day t do not enter any candidate. Values are stored at full precision.

The 25 columns are the candidate pool. Chapter 4 screens them for drift before any selection. The first 15 follow the course example and are measured in reais or in shares. The other 10 come from the literature. Nine of them are returns, ratios, or log changes, which do not grow with the price level. Amihud21 is measured per real traded. The adjusted close of WEG rises from about 0.09 reais in 2000 to about 55 reais in 2026, so a lag of the price mostly tracks that rise.

Definitions

The three price averages are exponentially weighted means of closes through day $t-1$. The half-life is τ, the smoothing weight is α, and observation i steps back has weight w_i:

$$\alpha = 1 - \exp(-\ln 2/\tau), \qquad w_i = (1-\alpha)^i.$$

CloseEMA2, CloseEMA4, and CloseEMA8 use $\tau = 1$, 2, and 4. The names follow the course example. The weights are normalized, which is the adjusted form of the moving average.

FEATURE	DEFINITION	ROLE
CloseLag1	C_{t-1}	Example. Yesterday's price level.
CloseLag2	C_{t-2}	Example. Price level two sessions back.
CloseLag3	C_{t-3}	Example. Price level three sessions back.
HMLLag1	$H_{t-1} - L_{t-1}$	Example. Yesterday's range, in reais.
HMLLag2	$H_{t-2} - L_{t-2}$	Example.
HMLLag3	$H_{t-3} - L_{t-3}$	Example.
OMCLag1	$O_{t-1} - C_{t-1}$	Example. Yesterday's open-to-close change, in reais.
OMCLag2	$O_{t-2} - C_{t-2}$	Example.
OMCLag3	$O_{t-3} - C_{t-3}$	Example.
VolumeLag1	V_{t-1}	Example. Undefined when the recorded volume is zero.
VolumeLag2	V_{t-2}	Example. Same zero-volume rule.
VolumeLag3	V_{t-3}	Example. Same zero-volume rule.
CloseEMA2	weighted mean above, $\tau = 1$	Example.
CloseEMA4	weighted mean above, $\tau = 2$	Example.
CloseEMA8	weighted mean above, $\tau = 4$	Example.
Ret1	$\log(C_{t-1}/C_{t-2})$	One-day reversal (Jegadeesh, 1990; Lehmann, 1990).
Ret21	$\log(C_{t-1}/C_{t-22})$	About one month of past return.
Mom12_1	$\log(C_{t-21}/C_{t-252})$	Twelve-month return, skipping the most recent month (Jegadeesh and Titman, 1993).
RVol21	$\sqrt{\frac{1}{20}\sum_{s=t-21}^{t-1}(r_s - \bar{r})^2}$	Recent volatility. Sample standard deviation of 21 daily log returns.

FEATURE	DEFINITION	ROLE
Parkinson21	$\frac{1}{21}\sum_{s=t-21}^{t-1}\frac{[\log(H_s/L_s)]^2}{4\ln 2}$	Range-based variance (Parkinson, 1980). It uses the high and the low.
LogVolChg1	$\log(V_{t-1}/V_{t-2})$	Volume shock, on positive volume (Campbell, Grossman, and Wang, 1993).
Amihud21	$\frac{1}{21}\sum_{s=t-21}^{t-1}\frac{\lvert r_s\rvert}{C_sV_s}$	Illiquidity (Amihud, 2002). Each session in the window must have positive volume.
Range1	$(H_{t-1}-L_{t-1})/C_{t-1}$	Yesterday's range as a fraction of price. Scale-free HML.
Intraday1	$\log(C_{t-1}/O_{t-1})$	Yesterday's open-to-close return. Scale-free OMC.
Overnight1	$\log(O_{t-1}/C_{t-2})$	Yesterday's overnight gap.

Volume and the estimation sample

Yahoo records volume of zero on 1,814 of the 6,721 downloaded sessions, including nearly every session from 2000 through 2006. A zero is treated as missing volume, not as a session with no shares traded. Volume lags, the log volume change, and the Amihud ratio are undefined on a zero print. A row is kept only when all 25 candidates are defined. Closes from before the first kept row still enter the moving averages and Mom12_1. They are history, not rows of their own. No missing value is filled in.

The kept table has 4,418 sessions, from 3 July 2007 through 9 October 2026. Of these, 2,229 (50.5 percent) are up days and 2,189 are flat or down. Within that span, 368 downloaded sessions are absent because a zero-volume print leaves at least one candidate undefined.

The same file stores the day's raw prices, the same-day range, the same-day open-minus-close, and the log return, so the candidates can be checked. Those columns are not part of the 25.

RSI, MACD, Bollinger bands, fractional differentiation, and any series that is not in this file of daily open, high, low, close, and volume are outside the candidate list. Chapter 2 cuts this table into a development window and a hold-out window.

4. Observable universe

A classifier fit on earlier sessions is applied to later ones. That works only for a column whose later values fall in the range seen earlier. A tree cannot extrapolate: a value above everything in the training rows lands in the same leaf as the largest training value. A column measured in reais or in shares grows with the company, so its later values leave the earlier range and the column marks the calendar period instead of the next session.

The screen reads the development file only. It splits development into thirds of 1,241 sessions. The first third runs from 2007-07-03 through 2013-03-04. The last third runs from 2018-11-28 through 2023-12-28. For each candidate x, let $q_{0.01}$ and $q_{0.99}$ be the 1st and 99th percentiles over the first third. The outside share is

$$s = \frac{1}{n_3} \sum_{t \in \text{last third}} \mathbf{1}\{x_t < q_{0.01} \text{ or } x_t > q_{0.99}\}.$$

A column with a stable distribution leaves about 2 percent of the last third outside that range. A candidate enters the observable universe when $s \le 0.1$, five times that rate.

FEATURE	FIRST-THIRD MEDIAN	LAST-THIRD MEDIAN	OUTSIDE SHARE	SCREEN
CloseLag1	1.96	29.36	100.0%	Excluded
CloseLag2	1.959	29.36	100.0%	Excluded
CloseLag3	1.959	29.36	100.0%	Excluded
HMLLag1	0.05536	0.7427	91.5%	Excluded
HMLLag2	0.05522	0.7427	91.5%	Excluded
HMLLag3	0.05504	0.7427	91.9%	Excluded
OMCLag1	9.92×10^{-4}	0.008939	74.3%	Excluded
OMCLag2	9.86×10^{-4}	0.008939	74.5%	Excluded
OMCLag3	9.72×10^{-4}	0.008939	74.5%	Excluded
VolumeLag1	2.05×10^{6}	7.55×10^{6}	16.2%	Excluded
VolumeLag2	2.04×10^{6}	7.55×10^{6}	16.2%	Excluded
VolumeLag3	2.04×10^{6}	7.55×10^{6}	18.4%	Excluded
CloseEMA2	1.957	29.42	100.0%	Excluded
CloseEMA4	1.956	29.43	100.0%	Excluded
CloseEMA8	1.962	29.46	100.0%	Excluded
Ret1	0	4.93×10^{-4}	2.4%	Kept
Ret21	0.0134	0.01481	6.5%	Kept
Mom12_1	0.1359	0.2148	8.9%	Kept
RVol21	0.01975	0.01929	2.3%	Kept
Parkinson21	4.32×10^{-4}	3.55×10^{-4}	6.0%	Kept
LogVolChg1	4.70×10^{-4}	-0.017	0.0%	Kept
Amihud21	4.60×10^{-9}	6.60×10^{-11}	100.0%	Excluded
Range1	0.02895	0.02783	1.1%	Kept

FEATURE	FIRST-THIRD MEDIAN	LAST-THIRD MEDIAN	OUTSIDE SHARE	SCREEN
Intraday1	-5.15×10^{-4}	-5.40×10^{-4}	2.2%	Kept
Overnight1	0.001009	0.001433	2.0%	Kept

9 of 25 candidates pass: Ret1, Ret21, Mom12_1, RVol21, Parkinson21, LogVolChg1, Range1, Intraday1, Overnight1. The 16 excluded columns are CloseLag1, CloseLag2, CloseLag3, HMLLag1, HMLLag2, HMLLag3, OMCLag1, OMCLag2, OMCLag3, VolumeLag1, VolumeLag2, VolumeLag3, CloseEMA2, CloseEMA4, CloseEMA8, Amihud21. Every excluded column is measured in reais, in shares, or, for Amihud21, per real traded. Each scale-free counterpart that measures the same quantity passes the screen: Ret1 for the lagged close, Range1 for high minus low, Intraday1 for open minus close, and LogVolChg1 for volume. The limit of 10 percent and the two percentiles are the only settings of the screen.

5. Feature selection

Selection uses the 9 columns from the previous chapter and the 3,723 development sessions. The hold-out file is not read.

How a subset is scored

Each subset is fit by logistic regression and scored on sessions the fit did not see. Development is cut into the same 5 time-series folds used later for XGBoost, with a gap of 10 sessions. In each fold the model is fit on earlier sessions and scored on the next block. The score is the log loss,

$$\ell = -\frac{1}{n}\sum_{t}\left[y_t \ln \hat{p}_t + (1 - y_t)\ln(1 - \hat{p}_t)\right],$$

averaged over the 5 folds. Lower is better. A model with no column predicts the up share of its training block every day. Its log loss, 0.69361, is the baseline.

Forward search

The search starts with no column. At each step it adds the one column that lowers the log loss most. It stops when no column lowers it further. The gain is how far the log loss falls below the baseline, in thousandths.

STEP	ADDED	MEAN LOG LOSS	GAIN (×1000)
1	Ret1	0.69304	0.575
2	Intraday1	0.69303	0.581

The search stops at Ret1, Intraday1, with a gain of 0.581.

Noise gate

A search like this always finds some gain, even when the columns carry no information. The gate asks how large a gain noise alone produces.

The label column is shifted in time by at least 252 sessions, so each day's features line up with the label of another day. The forward search is then run again on the shifted label. This is repeated 200 times. The gains from these runs are what noise alone produces. Their median is 0.143, and 95 percent of them are below 1.057. A selection is kept only if its gain is above that line.

The actual gain, 0.581, is below that line. 19 percent of the shifted runs reach a gain at least as large. The search therefore does not single out any column, and the next chapter fits XGBoost on all 9 columns.

A fair estimate

The columns above were chosen with the whole development window. Scoring them on the same window would be too generous. To get a fair estimate, the full procedure, search and gate, is repeated inside each fold using only that fold's training sessions. When the gate fails, the fold uses all 9 columns. The resulting logistic regression is scored on the next block, which the procedure did not see.

FOLD	SCORED BLOCK	FORWARD PICK	GATE PASSED	COLUMNS USED	ROC AUC	GAIN (×1000)
1	2010-04-19 to 2013-03-06	Ret1	No	all 9	0.4782	-17.347
2	2013-03-07 to 2015-11-04	Ret1	No	all 9	0.4869	-0.781
3	2015-11-05 to 2018-11-28	none	No	all 9	0.5507	2.636
4	2018-11-29 to 2021-06-04	Ret1	No	all 9	0.5127	-8.595
5	2021-06-07 to 2023-12-28	Ret1	No	all 9	0.5063	0.620

The gate passes in 0 of 5 folds. The mean ROC AUC is 0.5070, and the mean gain is -4.693. A negative gain means the model did worse than predicting the training up share.

6. XGBoost

The classifier uses the 9 columns passed on by the previous chapter: Ret1, Ret21, Mom12_1, RVol21, Parkinson21, LogVolChg1, Range1, Intraday1, Overnight1. Optuna runs 100 trials on the 3,723 development sessions. Each trial is a different combination of tree depth, minimum child weight, row subsample, learning rate, and number of trees. The next trial is aimed near combinations that already scored well. The score is the mean ROC AUC across 5 time-series folds, with a gap of 10 sessions:

$$\overline{\mathrm{AUC}} = \frac{1}{5}\sum_{k=1}^{5} \mathrm{AUC}_k.$$

Tree depth runs from 1 to 6. The subsample is kept between 0.5 and 1. The seed is 21. The hold-out file is not read.

Many trials score almost the same, so the single best one is partly luck. The search keeps the simplest trial whose score is within one standard error of the best: shallower trees first, then fewer trees. The best mean AUC is 0.5115, and 83 of the 100 trials fall within one standard error of it.

The chosen settings are depth 1, minimum child weight 3, subsample 0.782, learning rate 0.007, and 51 trees. Their mean AUC is 0.5077. A logistic regression on the same columns and folds reaches 0.5070.

Ten best trials

RANK	MEAN AUC	DEPTH	MIN CHILD	SUBSAMPLE	LEARNING RATE	TREES
1	0.5115	1	1	0.677	0.006	193
2	0.5108	1	1	0.697	0.005	182
3	0.5107	1	3	0.678	0.007	171
4	0.5106	1	3	0.705	0.008	169
5	0.5105	1	1	0.724	0.006	76
6	0.5105	1	1	0.681	0.006	161
7	0.5105	1	2	0.729	0.006	187
8	0.5105	1	3	0.697	0.005	197
9	0.5103	1	2	0.673	0.005	53
10	0.5103	1	1	0.924	0.007	399

7. Hold-out

The classifier is fit once on the 3,723 development sessions, from 2007-07-03 through 2023-12-28. The settings are the ones chosen in the previous chapter: depth 1, minimum child weight 3, subsample 0.782, learning rate 0.007, and 51 trees. The columns are Ret1, Ret21, Mom12_1, RVol21, Parkinson21, LogVolChg1, Range1, Intraday1, Overnight1. The seed is 21. The fit does not see the hold-out.

The score uses the 685 hold-out sessions, from 2024-01-16 through 2026-10-09. The 10 sessions from 2024-01-02 through 2024-01-15 stay out of the fit and out of the score. A session is called up when the fitted probability satisfies

$$\hat{y}_t = \mathbf{1}\{\hat{p}_t \geq 1/2\}.$$

Three reference rules are scored with it. The logistic regression uses the same columns and has no tuned setting. Always up predicts an up day every session. Repeat previous direction predicts that today's direction equals the direction of the immediately previous session. For the first hold-out day, that previous session is 2024-01-15.

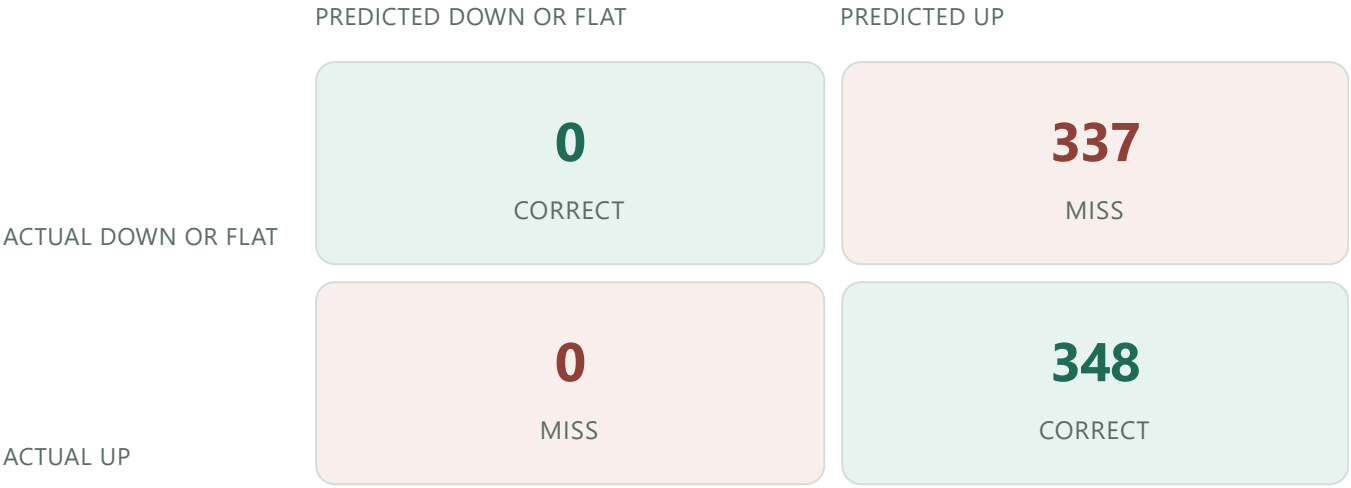
Down hit counts down-or-flat days predicted as down-or-flat. Down miss counts those days predicted as up. Up miss counts up days predicted as down-or-flat. Up hit counts up days predicted as up. With up as the positive class,

$$\mathrm{TPR} = \frac{\text{up hit}}{\text{up hit} + \text{up miss}}, \qquad \mathrm{FPR} = \frac{\text{down miss}}{\text{down miss} + \text{down hit}}.$$

The always-up rule assigns the same score to every session, so its ROC AUC is 0.5.

Hold-out scores

RULE	ACCURACY	ROC AUC	DOWN HIT	DOWN MISS	UP MISS	UP HIT
XGBoost	0.5080	0.5303	0	337	0	348
Logistic regression	0.5226	0.5267	172	165	162	186
Always up	0.5080	0.5000	0	337	0	348
Repeat previous direction	0.4599	0.4597	152	185	185	163



Hold-out calls for the fitted classifier. A green cell is a correct call.

The hold-out share of up days is 0.5080. Always up is therefore right on 0.5080 of the hold-out sessions. XGBoost is right on 0.5080, with ROC AUC 0.5303. All its fitted probabilities are above one half, so it calls every session up, like always up. The logistic regression is right on 0.5226, with ROC AUC 0.5267. Repeat previous direction is right on 0.4599.

Chance or skill

With 685 sessions, part of any difference is chance. The hold-out is resampled 2,000 times in blocks of 20 consecutive sessions, and each rule is scored again on every sample. The middle 95 percent of those scores is the range below. The classifier counts as skilled only if its whole ROC AUC range sits above 0.5. That rule was set before the hold-out was scored. Log loss measures the fitted probabilities directly; lower is better.

RULE	ROC AUC	95% RANGE	LOG LOSS
XGBoost	0.5303	0.4921 to 0.5729	0.6929
Logistic regression	0.5267	0.4731 to 0.5858	0.6922
Always up	0.5000	0.5000 to 0.5000	0.6930
Repeat previous direction	0.4597	0.4160 to 0.5020	n/a

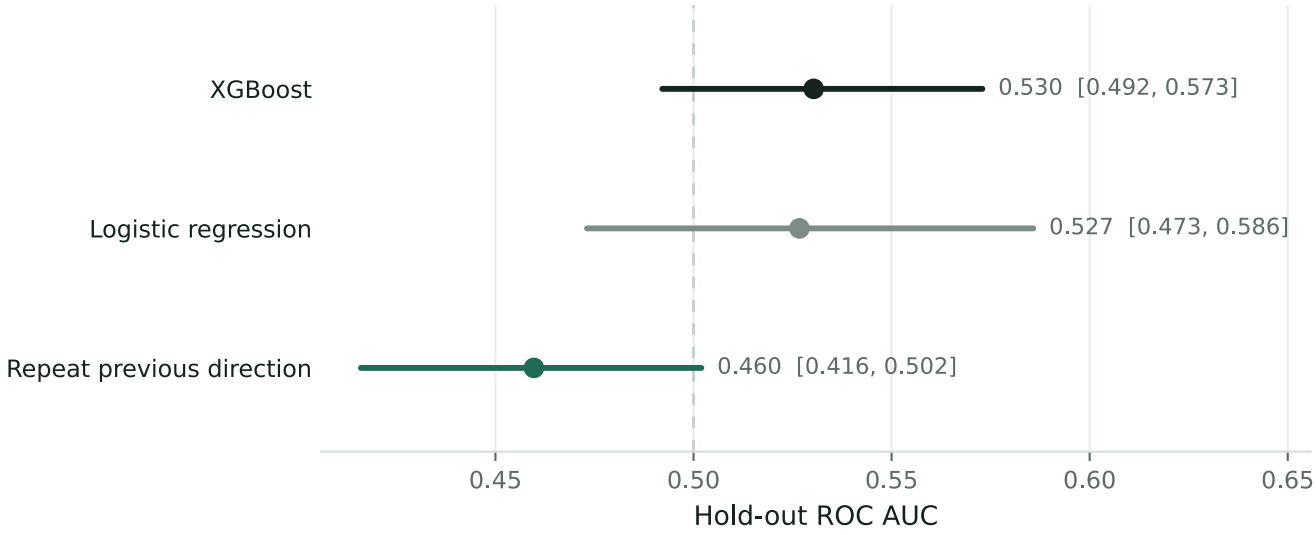


Figure 1. Hold-out ROC AUC with its 95 percent range. The dashed line is 0.5.

The XGBoost range runs from 0.4921 to 0.5729. It includes 0.5, so the classifier shows no skill on the hold-out. With 685 sessions, a range clears 0.5 only when the ROC AUC is near 0.54 or higher.

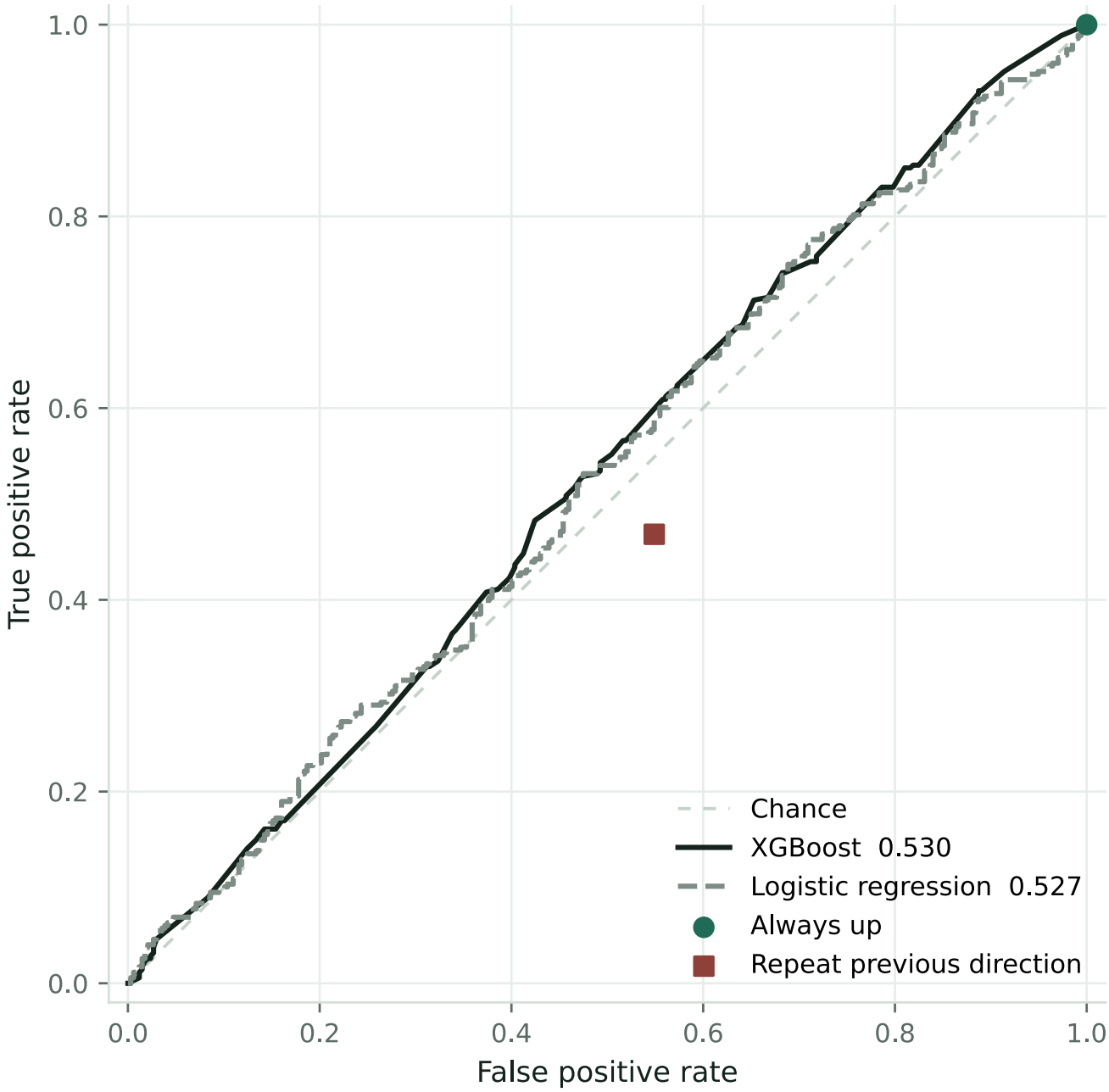


Figure 2. Hold-out ROC. The curves stay close to the diagonal.

8. Conclusion

The drift screen keeps 9 of the 25 candidates. The 16 it removes are measured in reais, in shares, or per real traded. They follow the growth of the company, not the next session.

On those 9 columns, no subset beats noise. The forward search picks Ret1 and Intraday1, but 19 percent of the runs on shifted labels gain at least as much. XGBoost therefore uses all 9 columns. Its mean ROC AUC on the development folds is 0.5077, and the fair estimate from the nested folds is 0.5070.

On the 685 hold-out sessions XGBoost has ROC AUC 0.5303 and accuracy 0.5080, the same accuracy as always up. Its 95 percent range for ROC AUC, 0.4921 to 0.5729, includes 0.5. The logistic regression has ROC

AUC 0.5267 and accuracy 0.5226, and its range also includes 0.5. Neither classifier shows skill. In this design, the next-session direction of WEGE3.SA is not predictable from daily open, high, low, close, and volume.

The course example reports classification accuracy of 81.8 percent and an AUC of 0.82 after fitting and scoring the same rows. Its time-series cross-validation accuracy was about 50 to 51 percent, and the example states that training-set predictions are better than should be expected on new data. The hold-out numbers in this report sit with that cross-validation result.

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Software

Prices were downloaded with yfinance. Candidate columns were built with Polars. The forward search checks its logistic fit against scikit-learn's unpenalized logistic regression. The classifier is XGBoost, and the hyperparameter search is Optuna. The figures use Matplotlib.